

To evaluate the proposed A/B tests, I conducted individual interviews for each feature. After each feature, I will present the average results from the users' responses, including feedback from a special guest participant.

After each interview task, participants were asked to rate both Version A and Version B out of 5 based on clarity, usefulness, motivation, trust, and overall preference.

Gamification Features

A/B Test 1 - Visible Reward Points Progress

Item	Description
Feature	Users collect reward points from purchasing products, and they can track their progress toward rewards from the profile page.
Version A - Control	No points are shown (no pointing system)
Version B - Variation	A reward points system is added. The profile page shows a progress bar, such as "320 points collected — 180 points left to unlock 10% off." Product cards also show how many points the user can earn from each product, for example "Earn 15 points."
Independent variable	Adding a visible reward points system with a progress bar on the profile page and points displayed on product cards.
Primary metric	Repeat purchase rate.
Secondary metrics	Add-to-cart rate, order completion rate, average order value, profile page visits, 7-day retention.
Hypo	Adding a visible reward points system with a profile progress bar and product points will increase repeat purchases because users can see the value gained from each product and track their progress toward a reward.
Expected outcome	Version B is expected to perform better because showing points on products makes the reward immediate, while the progress bar on the profile page motivates users to continue purchasing until they unlock a benefit.

Average User Feedback

Most participants preferred Version B because the reward points were visible and easy to notice. Users said that seeing points on product cards made the shopping experience feel more rewarding. The profile progress bar also helped users understand how close they were to receiving a discount.

Version A was considered simple, but users did not feel motivated to return because there was no visible reward system.

Version	Average Rating / 5
Version A — Control	2.8 / 5
Version B — Variation	4.5 / 5

Result Interpretation

Version B performed better because it made the reward system clearer and more motivating. The feature is expected to improve repeat purchases and encourage users to add more products to their cart.

A/B Test 2 - Reward After First Order

Item	Description
Feature	Users receive a reward after completing their first order from the app.
Version A - Control	The current app shows the normal order confirmation screen after checkout, with no reward, badge, or achievement after the first order.
Version B - Variation	After completing the first order, the app shows a reward message or achievement badge, such as “First Order Completed — You earned 50 points” or “Welcome Reward Unlocked.”
Independent variable	Adding a first-order reward after checkout.
Primary metric	Second purchase rate.

Secondary metrics	Repeat purchase rate, 7-day retention, reward interaction rate, order completion satisfaction rating.
Hypo	Adding a reward after the first order will increase the second purchase rate because users receive positive feedback and have a clear incentive to return and order again.

Average User Feedback

Participants found **Version B** more engaging because the reward message made the first order feel more valuable. The “First Order Completed” or “Welcome Reward Unlocked” message created a positive feeling after checkout.

Version A was clear as a normal confirmation screen, but users said it felt basic and did not give them a reason to return.

Version	Average Rating / 5
Version A — Control	3.0 / 5
Version B — Variation	4.5 / 5

Result Interpretation

Version B performed better because it added positive feedback after the first order. This can encourage users to place a second order and improve short-term retention.

A/B Test 3 - Free Shipping Spending Progress Bar

Item	Description
Feature	Users see how much more they need to spend to unlock free shipping.
Version A - Control	The cart shows the normal subtotal, delivery fee, and checkout button without a free shipping progress indicator.
Version B - Variation	The cart includes a progress bar showing how close the user is to free shipping, for example: “Spend 120 EGP more to unlock free shipping” when the free shipping threshold is 500 EGP.
Independent variable	Adding a free shipping progress bar in the cart.
Primary metric	Average order value.

Secondary metrics	Order completion rate, add-to-cart rate, cart abandonment rate, free shipping achievement rate.
Hypo	Adding a free shipping progress bar will increase average order value because users will be motivated to add more items when they see they are close to unlocking free shipping.
Expected outcome	Version B is expected to perform better because the progress bar makes the benefit visible and encourages users to reach the spending goal.

Average User Feedback

Most participants strongly preferred **Version B**. They said the free shipping progress bar clearly explained how much more they needed to spend to avoid delivery fees. Several users mentioned that if the remaining amount was small, they would add another product to reach free shipping.

Version A was understandable, but users felt it gave less guidance during checkout.

Version	Average Rating / 5
Version A — Control	3.2 / 5
Version B — Variation	4.7 / 5

Result Interpretation

Version B is expected to perform better because it can increase average order value and reduce cart abandonment. However, product recommendations should be handled carefully in a pharmacy context.

Overall Gamification Feedback from Our Special Guest User: Dr. Amal

The doctor generally preferred Version B across the gamification features. She explained that if the app provides a clear benefit, such as discounts, reward points, or progress toward offers, she would be more likely to use the app more often. From her perspective, the gamification features made the experience more motivating because users could see a direct value from continuing to use the app. However, she also pointed out some layout and visual hierarchy issues that should be improved in the implementation milestone. These comments will be taken into consideration when refining the final design.

AR Features

A/B Test 1 - AR Medicine Category Label

Item	Description
Feature	After scanning the product barcode, the app shows the medicine category.
Version A - Control	The barcode scan shows the product name, price, availability, and normal product details.
Version B - Variation	The barcode scan shows an AR category label on top of the product/package, such as "Psychiatric Medicine," "Eye Care," "Pain Relief," "Antibiotic," or "Vitamins."
Independent variable	Showing the medicine category as an AR label after scanning.
Primary metric	Product understanding rate.
Secondary metrics	Product detail page visits, time spent on scan result, add-to-cart rate, user confidence rating.
Hypo	Showing the medicine category after barcode scanning will improve users' understanding of the product because the category label gives users immediate context about what the medicine is used for.
Expected outcome	Version B is expected to perform better because users can quickly understand the type of medicine without reading long product descriptions.

Average User Feedback

Participants preferred **Version B** because the AR category label helped them quickly understand the type of medicine after scanning the barcode. Users said that labels such as "Eye Care," "Pain Relief," or "Vitamins" made the scan result easier to understand without reading long details.

Version A still provided useful information, but users needed more time to understand the product category.

Version	Average Rating / 5
Version A — Control	3.1 / 5
Version B — Variation	4.6 / 5

Result Interpretation

Version B performed better because it improved product understanding and user confidence. This feature is especially useful for users who scan unfamiliar products.

A/B Test 2 - AR Usage Animation Button

Item	Description
Feature	After scanning the product barcode, users can press a button to view an animation explaining how the medicine is used.
Version A - Control	The product page shows written usage instructions only.
Version B - Variation	The scan result includes a button such as “Show How to Use,” which opens a short AR animation, for example how to apply eye drops, use a spray, take syrup, or use an inhaler.
Independent variable	Adding an AR usage animation button.
Primary metric	Instruction comprehension rate.
Secondary metrics	Button click-through rate, animation completion rate, time on task, user confidence rating, support/help clicks.
Hypo	Adding an AR usage animation button will improve users’ understanding of how to use the medicine because visual step-by-step guidance is easier to follow than text instructions.
Expected outcome	Version B is expected to perform better, especially for products that require correct physical usage, such as eye drops, inhalers, sprays, and syrups.

Average User Feedback

This was one of the strongest-performing features. Participants preferred **Version B** because the “Show How to Use” button made the instructions feel easier to understand. Users said that an animation would be especially helpful for products such as inhalers, sprays, syrups, and eye drops.

Version A was considered acceptable for simple medicine, but less effective for products that require a specific physical usage method.

Version	Average Rating / 5
Version A — Control	3.2 / 5
Version B — Variation	4.8 / 5

Result Interpretation

Version B is expected to perform significantly better because it improves instruction comprehension and user confidence. This feature has strong value in a pharmacy app because it supports safer medicine usage.

A/B Test 3 - AR Origin Indicator (Imported or Egyptian)

Item	Description
Feature	After scanning the product barcode, the app shows whether the product is imported or made in Egypt.
Version A - Control	The barcode scan shows normal product details without highlighting product origin.
Version B - Variation	The scan result shows a small AR object or badge beside the product, such as an Egypt flag for local products or a globe/import icon for imported products.
Independent variable	Showing product origin as an AR visual badge.
Primary metric	Product trust/clarity rating.
Secondary metrics	Add-to-cart rate, product comparison clicks, time spent on scan result, user satisfaction rating.
Hypo	Showing whether the product is imported or Egyptian will improve user trust and clarity because some users care about product origin when choosing between alternatives.
Expected outcome	Version B is expected to perform better because users can quickly understand product origin without searching through details.

Average User Feedback

The feedback for **Version B** was mixed. Around **50% of users** thought the AR origin indicator was useful because it made it easier to quickly identify whether the product was imported or made in Egypt. These users felt that product origin could help them compare between similar products and make the product information feel more transparent.

However, the other **50% of users** thought the feature was unnecessary and said they would probably not use it in the future. They explained that product origin is useful information, but they would prefer it to be shown as normal written text in the product details rather than as an AR badge or visual object. For them, using AR for origin information felt like more effort than needed.

Version	Average Rating / 5
Version A — Control	3.2 / 5
Version B — Variation	3.5 / 5

Result Interpretation

This test did not show a clear preference for Version B. While half of the users found the AR origin indicator useful, the other half felt it was unnecessary and preferred a simpler written format. Therefore, the AR origin indicator should not be prioritized compared to stronger features such as the AR usage animation or AR medicine category label.

A better recommendation would be to keep the origin information in the app, but display it as clear written text or a small simple badge inside the product details page rather than as a full AR feature.

Overall AR Feedback from Our Special Guest User: Dr. Amal

For the AR features, the doctor was even more interested in Version B. She described the AR concepts as creative, useful, and more engaging than the regular product information layout. She was especially interested in seeing the final implementation because the AR features could make medicine information easier to understand and more interactive for users. Overall, she was more satisfied with Version B because it added a clearer and more memorable experience compared to the control version.

Figma Deliverables:

WireFrame:

<https://www.figma.com/design/ejrsegQ4eT6wZOYcilOVp1/Elezaby-AB-version--Wireframe?node-id=0-1&t=MbDG2nddho10j3kO-1>

Mockup:

<https://www.figma.com/design/i7yPOMNXNIlcvmaFed4tz7/El-Ezaby-Interactive-Prototype?node-id=0-1&t=DUOfMcaJkfd5PBh4-1>

Pipeline:

The process started with creating the initial wireframe. Then, Claude was used to generate an HTML version of the intended mockup, which helped guide the visual structure and layout before moving into Figma. Finally, the full mockup was designed in Figma. Users found the final version more interesting, visually vibrant, and engaging to try.